NetApp Knowledge Base

How to decommission, repurpose, or reset an ONTAP 9 system?

https://kb.netapp.com/on-prem/ontap/Ontap_OS/OS-KBs/How_to_decommission_repurpose_or_reset_…
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Applies to

- ONTAP 9
- Repurpose ONTAP cluster
- Performing a full system reset to factory default settings

Description

When to Repurpose

Whenever an HA pair running ONTAP needs to be repurposed or reused, the configuration and data must be

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wiped from the system for it to be prepared to be utilized again.

Warning: All data and configuration for the HA pair will be removed during this procedure. Prior to proceeding with this repurposing process, move all data to another ONTAP cluster.

Procedure

Preparation

- This procedure assumes an HA pair (two nodes) are being repurposed. They are referred to as node1 and node2 within this procedure.
- Repurposing the HA pair indicates an ONTAP configuration previously existing on the nodes and hardware was properly installed according to ONTAP hardware systems documentation.
- If more than two nodes exist in the cluster, the HA pair must first be removed from the cluster.
- This process does not apply to ONTAP systems with SnapLock Compliance enabled as that prevents any reinitialize disk operation.
- This process does not apply to ASA r2 systems, for those platforms follow How to reinitialize ASA r2 systems?
- For MetroCluster environments, refer to How to remove nodes from a MetroCluster configuration - Resolution Guide.
- One of the requirements is understanding if the HA pair utilizes FIPS-certified SAS or NVMe drives (FIPS drives).
    - Refer to How to tell if a drive is FIPS certified
    - FIPS certificate information for FIPS drives can be found on the Disk Drive & Firmware Matrix.
- Decide if you wish to use partitions (supported for both AFF and FAS models starting in ONTAP 9.2) or whole disks for the root aggregate and refer to Advanced Drive Partitioning (ADP). Refer to Hardware Universe for platform and ONTAP version support for "ADP Root Partition Configuration". This procedure assumes partitioned disks will be used.

Halt Both Nodes

1. Login via local console cable to node1 and node2. Alternatively connect to SP/BMC over SSH and enter the "`system console`" command on each node.
2. Is the HA pair using FIPS drives or self-encrypting NVMe drives (SED drives)?
    a. If **yes**, then you **MUST** follow the process of Returning a FIPS drive or SED to unprotected mode for all disks within the HA pair. Failure to do this may result in future data loss.

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   b. If **no**, consider How to sanitize all disks in ONTAP 9.6+ or proceed directly to the next step.

3. If the prompt currently reads "LOADER", the system is currently halted. Otherwise, login via ONTAP and halt both nodes and then each node will halt and reach the "LOADER" prompt.

```
cluster1::> system node halt -node * -skip-lif-migration-before-shutdown
true -ignore-quorum-warnings true -inhibit-takeover true

Warning: Are you sure you want to halt node "cluster1-01"?
{y|n}: y

Warning: Are you sure you want to halt node "cluster1-02"?
{y|n}: y
```

Configure Environment Variables

If the system is being decommissioned, skip to Sanitize FIPS or SED Drives and Set HA State.

4. Otherwise perform the following commands on both node1 and node2.
5. Capture the current environment variables.

```
LOADER-A> printenv
```

6. Clear the current non-default variables.

```
LOADER-A> set-defaults
```

7. Is the HA pair using FIPS drives?
   a. If **yes**, set the value to enable FIPS encryption support for drives.

```
LOADER-A> setenv bootarg.storageencryption.support true
```

   b. If using self-encrypting NVMe drives (SED drives) or non-encrypting drives, unset the value to disable FIPS encryption support for drives.

```
LOADER-A> unsetenv bootarg.storageencryption.support
```

8. Set the date and time. **This must set in UTC time only.**

```
LOADER-A> set date mm/dd/yyyy

LOADER-A> set time hh:mm:ss

LOADER-A> show date
```

The date and time are given in UTC.

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9. Prevent the system from automatically booting into ONTAP. Save the environment variable changes and reboot the node for the changes to be recognized.

```
LOADER-A> setenv AUTOBOOT false

LOADER-A> saveenv

LOADER-A> bye
```

Install ONTAP Image

Note: If the currently installed version of ONTAP is the desired version for the new cluster, skip this section and proceed with the Set HA State section.

10. Select a version of ONTAP 9 from the Recommended ONTAP releases on the NetApp Support Site that matches a minimum version for the platform model specified in Hardware Universe and download the corresponding image.
11. There are two methods available for installing ONTAP on a repurposed system.

    Option A: Netboot Process

    - Requires network connectivity
    - Requires a web server
    - Can be performed remotely through the SP/BMC

    Option B: Boot Recovery Process

    - Requires a USB flash device
    - Requires **no** network connectivity
    - Requires on site presence

12. Follow the appropriate steps for the option selected in the previous step.

    - **Option A**: Follow How to perform the option 7 "Install new software first" from the ONTAP boot menu for both node1 and node2.
    - **Option B**: Follow How to use the boot_recovery LOADER command for installing ONTAP for initial setup of a system for both node1 and node2.

13. At the end of the process, both node1 and node2 will be at the at the "LOADER" prompt instead of maintenance mode.

Sanitize FIPS or SED Drives and Set HA State

14. Perform the following commands on both node1 and node2.

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15. Boot into maintenance mode. Answer when prompted to continue.

```
LOADER-A> boot_ontap maint

Continue with boot? yes
```

16. Is the HA pair using FIPS drives or self-encrypting NVMe drives (SED drives)?
    a. If **yes**, encrypt sanitize all disks to make the data on the disks inaccessible and prepare for reuse.

    ```
    *> disk encrypt sanitize -all

    If you are sure you want to proceed, type confirmation code:

            kill data
    Enter code:  kill data
    ```

    b. If **no**, proceed to the next step.

If the system is being decommissioned skip to Remove Disk Partitions and Ownership.

17. Change the HA configuration to the appropriate value for your configuration. The available options are in the chart, below.

Configuration	HA State
Two-node HA pair	ha
Single node system	non-ha
Eight- or four-node MetroCluster FC configuration	mcc
Two-node MetroCluster FC configuration	mcc-2n
MetroCluster IP configuration	mccip

```
*> ha-config modify chassis <ha_state_from_table>
*> ha-config modify controller <ha_state_from_table>
```

18. Confirm the changes are successful.

```
*> ha-config show

Chassis HA configuration: ha
Controller HA configuration: ha
```

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19. Halt the node.

```
*> halt
```

Remove Disk Partitions and Ownership

20. Perform the following commands on both node1 and node2.
21. Set the system for automatically booting into ONTAP.

```
LOADER-A> setenv AUTOBOOT true
```

22. Save the environment variable changes.

```
LOADER-A> saveenv
```

23. Boot into the startup menu.

```
LOADER-A> boot_ontap menu
```

24. When the startup menu appears, select option **9**.

```
Selection (1-11)? 9
```

WARNING

Note: It is very important to perform each of the remaining steps in a serial fashion (do not do any portion in parallel). Wait for each step to fully complete before proceeding with the next step.

25. Perform the following steps on node1.
26. On node1, from the "Advanced Drive Partitioning Boot Menu Options" select option 9a.

```
Selection (9a-9f)?: 9a
```

27. On node1, respond to continue when prompted.

With ONTAP 9.11.1 or newer on a fixed release of Bug ID 1397366:

```
Do you want to abort this operation (yes/no)? no
```

With previous releases of ONTAP:

```
Do you still want to continue (yes/no)? yes
```

28. On node1, it is **very** important to wait for the process to complete. Wait for it to return to the "Advanced

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Drive Partitioning Boot Menu Options" menu before continuing.

29. Perform the following steps on node2.
30. On node2, from the "Advanced Drive Partitioning Boot Menu Options" select option 9a.

```
Selection (9a-9f)?: 9a
```

31. On node2, respond to continue when prompted.

With ONTAP 9.11.1 or newer on a fixed release of Bug ID 1397366:

```
Do you want to abort this operation (yes/no)? no
```

With previous releases of ONTAP:

```
Do you still want to continue (yes/no)? yes
```

32. On node2, it is **very** important to wait for the process to complete. Wait for it to return to the "Advanced Drive Partitioning Boot Menu Options" menu before continuing.

If the system is being decommissioned, skip to Proceed with Final Steps.

Wipe Configuration and Initialize Node

33. Perform the following steps on node1.
34. On node1, from the "Advanced Drive Partitioning Boot Menu Options" select option 9b.

```
Selection (9a-9f)?: 9b
```

35. On node1, respond to continue when prompted.

With ONTAP 9.11.1 or newer on a fixed release of Bug ID 1397366:

```
Has option (9a) been completed on all the nodes (yes/no)? yes

Do you still want to continue (yes/no)? yes
```

With previous releases of ONTAP:

```
Do you still want to continue (yes/no)? yes
```

36. On node1, wait for the process to complete and for the following to appear.

```
Welcome to the cluster setup wizard.
```

37. Once the previous step is complete, perform the following steps on node2.

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38. On node2, from the "Advanced Drive Partitioning Boot Menu Options" select option 9b.

```
Selection (9a-9f)?: 9b
```

39. On node2, respond to continue when prompted.

With ONTAP 9.11.1 or newer on a fixed release of Bug ID 1397366:

```
Has option (9a) been completed on all the nodes (yes/no)? yes


Do you still want to continue (yes/no)? yes
```

With previous releases of ONTAP:

```
Do you still want to continue (yes/no)? yes
```

40. On node2, wait for the process to complete and for the following to appear.

```
Welcome to the cluster setup wizard.
```

Proceed with Final Steps

- If the system is being **repurposed**:
  - Both node1 and node2 the nodes are ready to continue with the cluster setup procedure and all steps in this article are complete
- If the system is being **decommissioned**:
  - Remove any remaining configuration with the wipeconfig boot menu process
  - Follow the Wiping the Entire Configuration of the SP/BMC
  - Physically power off the systems and report to Netapp that the system is no longer in service (there is a detailed video on NetappTV)

Additional Information

- What are the best practices for relocating your ONTAP cluster
- Nodes will not boot, error: Unable to restore data access on encrypting disk
- Error when sanitizing Self-Encrypting Drive (SED) or FIPS NSE drives
- Cannot remove node XX because its storage encryption devices use authentication keys

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